MAT1001 Calculus for Engineers

Line Integrals

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Line Integrals of Scalar Valued Functions

DEFINITION If C is a smooth curve in 2-space or 3-space, then the ***line integral of f with respect to s along C*** is

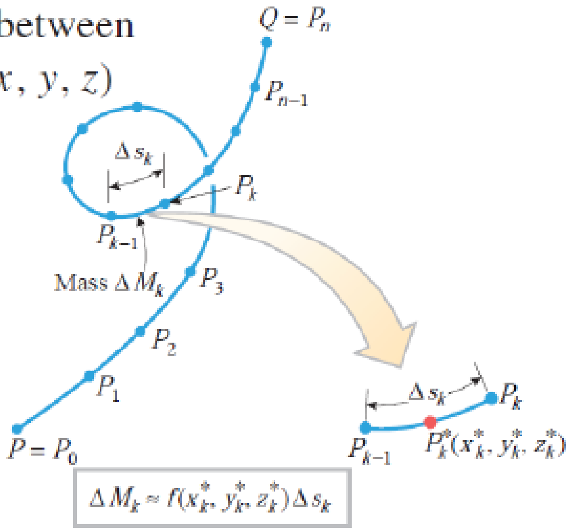
$$\int_C f(x, y)\, ds = \lim_{\max \Delta s_k \to 0} \sum_{k=1}^{n} f(x_k^*, y_k^*) \Delta s_k \qquad \boxed{\text{2-space}}$$

or

$$\int_C f(x, y, z)\, ds = \lim_{\max \Delta s_k \to 0} \sum_{k=1}^{n} f(x_k^*, y_k^*, z_k^*) \Delta s_k \qquad \boxed{\text{3-space}}$$

provided this limit exists and does not depend on the choice of partition or on the choice of sample points.

- **Example** We assume that we can model the wire by a smooth curve C between two points P and Q in 3-space. Given any point (x, y, z) on C, we let $f(x, y, z)$ denote the corresponding value of the density function.



To compute the mass of the wire,

- Divide C into n very small sections

$$P = P_0, P_1, P_2, \ldots, P_{n-1}, P_n = Q$$

- Δs_k be the length of the arc between P_{k-1} and P_k
- $P_k^*(x_k^*, y_k^*, z_k^*)$ be the arbitrary sampling point on the kth arc
- ΔM_k be the mass of the kth section

$$\Delta M_k \approx f(x_k^*, y_k^*, z_k^*)\Delta s_k$$

- The mass M of the entire wire can then be obtained by

$$M = \lim_{\max \Delta s_k \to 0} \sum_{k=1}^{n} f(x_k^*, y_k^*, z_k^*)\Delta s_k$$

- If C is a curve in 3-space that models a thin wire, and if $f(x, y, z)$ is the linear density function of the wire then the mass M of the wire is given by

$$M = \int_C f(x, y, z)\, ds$$

- **Example** If C is a smooth curve of arc length L, and f is identically 1, then

$$\int_C ds = \lim_{\max \Delta s_k \to 0} \sum_{k=1}^{n} \Delta s_k = \lim_{\max \Delta s_k \to 0} L = L$$

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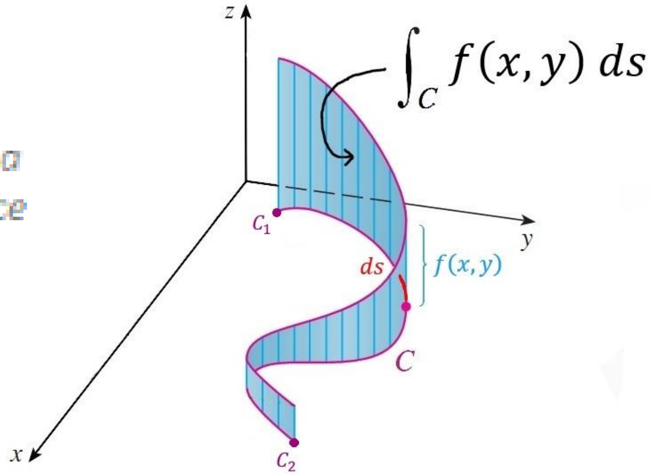
Line Integral - Area of a curved surface



Evaluation of Line Integral

DEFINITION If C is smoothly parametrized by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} \qquad (a \leq t \leq b)$$

then

$$\int_C f(x, y)\, ds = \int_a^b f(x(t), y(t))\|\mathbf{r}'(t)\|\, dt$$

Similarly, if C is a curve in 3-space that is smoothly parametrized by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} \qquad (a \leq t \leq b)$$

then

$$\int_C f(x, y, z)\, ds = \int_a^b f(x(t), y(t), z(t))\|\mathbf{r}'(t)\|\, dt$$

- **Example** Using the given parametrization, evaluate the line integral $\int_C (1 + xy^2)\, ds$.

(a) $C : \mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} \quad (0 \le t \le 1)$

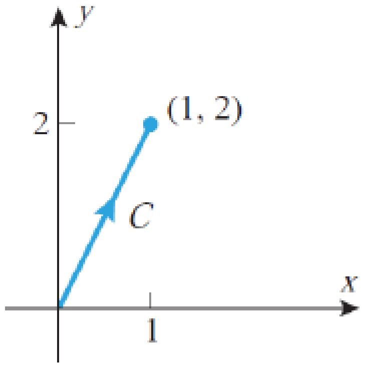
(b) $C : \mathbf{r}(t) = (1 - t)\mathbf{i} + (2 - 2t)\mathbf{j} \quad (0 \le t \le 1)$

Solution (a). Since $\mathbf{r}'(t) = \mathbf{i} + 2\mathbf{j}$, we have $\|\mathbf{r}'(t)\| = \sqrt{5}$ and it follows from Formula that

$$\int_C (1 + xy^2)\, ds = \int_0^1 [1 + t(2t)^2]\sqrt{5}\, dt$$

$$= \int_0^1 (1 + 4t^3)\sqrt{5}\, dt$$

$$= \sqrt{5}\left[t + t^4\right]_0^1 = 2\sqrt{5}$$



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- **Example** Using the given parametrization, evaluate the line integral $\int_C (1 + xy^2)\, ds$.

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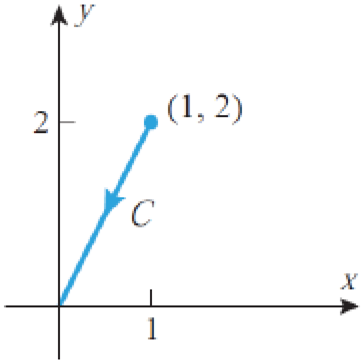
(b) $C : \mathbf{r}(t) = (1 - t)\mathbf{i} + (2 - 2t)\mathbf{j} \quad (0 \le t \le 1)$

Solution (b). Since $\mathbf{r}'(t) = -\mathbf{i} - 2\mathbf{j}$, we have $\|\mathbf{r}'(t)\| = \sqrt{5}$ and it follows from Formula that

$$\int_C (1 + xy^2)\, ds = \int_0^1 [1 + (1 - t)(2 - 2t)^2]\sqrt{5}\, dt$$

$$= \int_0^1 [1 + 4(1 - t)^3]\sqrt{5}\, dt$$

$$= \sqrt{5}\left[t - (1 - t)^4\right]_0^1 = 2\sqrt{5}$$



Evaluation of Line Integrals

If a curve C in the xy-plane that is given by parametric equations

$$x = x(t), \quad y = y(t) \qquad (a \leq t \leq b)$$

then we write (9) in the expanded form

$$\int_C f(x, y)\, ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

Similarly, if C is a curve in 3-space that is parametrized by

$$x = x(t), \quad y = y(t), \quad z = z(t) \qquad (a \leq t \leq b)$$

then we write (10) in the form

$$\int_C f(x, y, z)\, ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

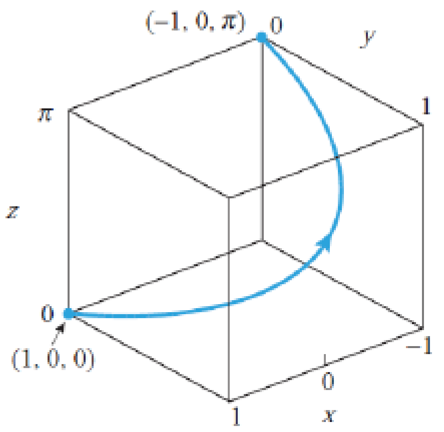
Example

▶ **Example** Evaluate the line integral $\int_C (xy + z^3)\, ds$ from $(1, 0, 0)$ to $(-1, 0, \pi)$ along the helix C that is represented by the parametric equations

$$x = \cos t, \quad y = \sin t, \quad z = t \qquad (0 \le t \le \pi)$$

Solution. From (12)

$$\begin{aligned}
\int_C (xy + z^3)\, ds &= \int_0^{\pi} (\cos t \sin t + t^3) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt \\
&= \int_0^{\pi} (\cos t \sin t + t^3) \sqrt{(-\sin t)^2 + (\cos t)^2 + 1}\, dt \\
&= \sqrt{2} \int_0^{\pi} (\cos t \sin t + t^3)\, dt \\
&= \sqrt{2} \left[\frac{\sin^2 t}{2} + \frac{t^4}{4} \right]_0^{\pi} = \frac{\sqrt{2}\pi^4}{4}
\end{aligned}$$



Special Case – Arc Length Parameter

In the special case where t is an arc length parameter, say $t = s$,

$$\int_C f(x, y)\, ds = \int_a^b f(x(s), y(s))\, ds \tag{13}$$

and

$$\int_C f(x, y, z)\, ds = \int_a^b f(x(s), y(s), z(s))\, ds \tag{14}$$

respectively.

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- **Example** Find the area of the surface extending upward from the circle $x^2 + y^2 = 1$ in the xy-plane to the parabolic cylinder $z = 1 - x^2$

Solution. It follows from (7) that the area A of the surface can be expressed as the line integral

$$A = \int_C (1 - x^2)\, ds \tag{15}$$

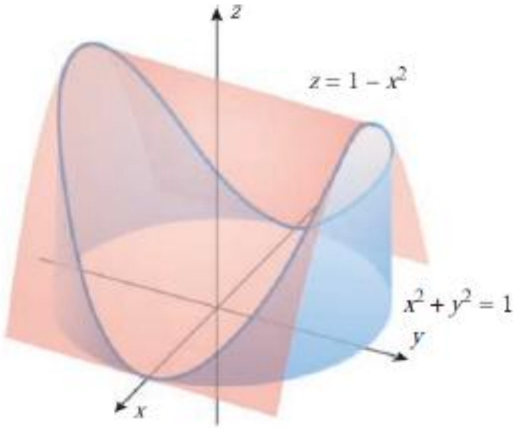
where C is the circle $x^2 + y^2 = 1$. This circle can be parametrized in terms of arc length as

$$x = \cos s, \quad y = \sin s \qquad (0 \le s \le 2\pi)$$

Thus, it follows from (13) and (15) that

$$A = \int_C (1 - x^2)\, ds = \int_0^{2\pi} (1 - \cos^2 s)\, ds$$

$$= \int_0^{2\pi} \sin^2 s\, ds = \frac{1}{2}\int_0^{2\pi} (1 - \cos 2s)\, ds = \pi$$



Line Integrals of Vector Fields

DEFINITION If $\mathbf{F}$ is a continuous vector field and C is a smooth oriented curve, then the ***line integral of*** $\mathbf{F}$ ***along*** C is

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

suppose that C is an oriented curve in the plane given in vector form by

$$\mathbf{r} = \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} \qquad (a \le t \le b)$$

If we write

$$\mathbf{F}(\mathbf{r}(t)) = f(x(t), y(t))\mathbf{i} + g(x(t), y(t))\mathbf{j}$$

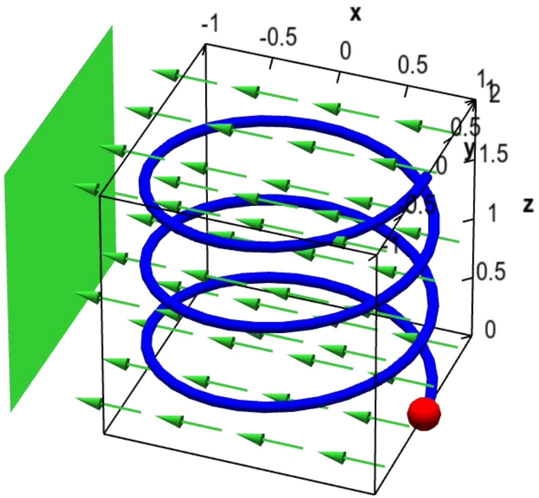
then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt$$

Formula is also valid for oriented curves in 3-space.

DEFINITION Let $\mathbf{F}$ be a vector field with continuous components defined along a smooth curve C parametrized by $\mathbf{r}(t)$, $a \leq t \leq b$. Then the **line integral of F along C** is

$$\int_C \mathbf{F} \cdot \mathbf{T}\, ds = \int_C \left(\mathbf{F} \cdot \frac{d\mathbf{r}}{ds} \right) ds = \int_C \mathbf{F} \cdot d\mathbf{r}.$$

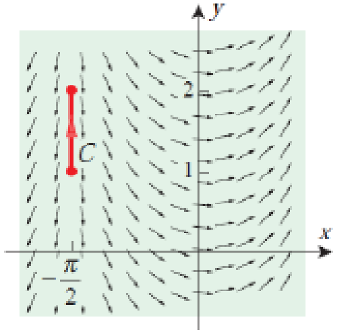


- **Example** Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y) = \cos x\mathbf{i} + \sin x\mathbf{j}$ and where C is the given oriented curve.

(a) $C : \mathbf{r}(t) = -\frac{\pi}{2}\mathbf{i} + t\mathbf{j} \quad (1 \leq t \leq 2)$

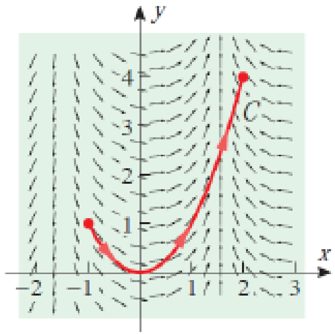
(b) $C : \mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} \quad (-1 \leq t \leq 2)$

Solution (a). Using (29) we have



$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_1^2 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt = \int_1^2 (-\mathbf{j}) \cdot \mathbf{j}\, dt = \int_1^2 (-1)\, dt = -1$$

Solution (b). Using (29) we have



$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{-1}^2 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt = \int_{-1}^2 (\cos t\mathbf{i} + \sin t\mathbf{j}) \cdot (\mathbf{i} + 2t\mathbf{j})\, dt$$

$$= \int_{-1}^2 (\cos t + 2t \sin t)\, dt = (-2t \cos t + 3 \sin t)\Big]_{-1}^2$$

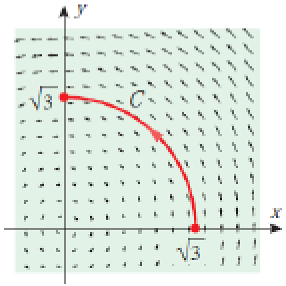
$$= -2\cos 1 - 4\cos 2 + 3(\sin 1 + \sin 2) \approx 5.83629$$

15.2.3 DEFINITION Suppose that under the influence of a continuous force field $\mathbf{F}$ a particle moves along a smooth curve C and that C is oriented in the direction of motion of the particle. Then the ***work performed by the force field*** on the particle is

$$\int_C \mathbf{F} \cdot d\mathbf{r} \tag{34}$$

- **Example** Determine the work done by a force $\mathbf{F}(x, y) = -y\mathbf{i} + x\mathbf{j}$ acting on a particle moving along the circle $x^2 + y^2 = 3$ from $(\sqrt{3}, 0)$ to $(0, \sqrt{3})$. Suppose that force is measured in pounds and distance is measured in feet.

Solution. $3\pi/2$ foot-pounds.



Thank You…